COMPLIANCE CHECKLIST

E-01 — Energy Performance

Riyadh Marina Tower

Not Started | 0 / 5–15 pts | Keystone | E · Energy | Manual pp.100–107

RATING SYSTEM
Mostadam commercial D+C (2019)

PROJECT STAGE
Design

TARGET LEVEL
Gold

GENERATED
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> Generated from the project's records in NetZero at the date and time above: the values recorded against each requirement, the files attached against each required document, and the AI's advisory read of those files. Nothing on this document was typed in by hand. A document reads Provided only when a file is attached against it. This is not a log of dated site inspections; where the manual asks for one, it is among the required documents below and the contractor's own sheets are what is attached there.

AIM

To promote informed decisions on the selection and installation of building components that achieve improved levels of energy efficiency and reduced carbon emissions.

REQUIRED DOCUMENTS AT A GLANCE

0	0	0	0	0
DOCUMENTS LISTED	DUE AT THIS STAGE	PROVIDED	WITH GAPS (AI)	MISSING

Requirements and required documents

Project at Design stage

Either/or: no path chosen yet

The project has not yet picked which of these options it will pursue, so their documents are not listed:

- #1 Prescriptive Option
- #2 Performance Option

What the manual requires the plan to contain

From the credit's supporting guidance (manual pp.100–107).

Performance Option

To determine the number of points achieved under this credit, a suitably Qualified Professional must develop a detailed energy model of two buildings:

- [ ] Baseline building as prescribed by:
  - [ ] ASHRAE Standard 90.1-2013, Appendix G
- [ ] Proposed building based on the design specifications and drawings

Additional Clarifications

- [ ] The proposed and baseline simulations must use the same energy modelling software and the same weather data file. The energy modelling software must meet the minimum requirements of ASHRAE Standard 90.1-2013, Appendix G (Section G2.2).
- [ ] The simulation program shall be capable of performing design load calculations to determine required HVAC equipment capacities and air and water flow rates in accordance with generally 101 accepted engineering standards and handbooks for both the proposed design and baseline building.
- [ ] Compliance with mandatory provisions of ASHRAE 90.1-2007 can be followed. However, overall comparison between proposed and baseline buildings shall be demonstrated against ASHRAE 90.1 2013.
- [ ] It is important to consider the correct climate zone relevant to project location as per the list of zones defined in ASHRAE Standards. KSA generally is categorized under Climate Zone 1 which is “Very Hot”. However, the sub-zone must be accurately identified depending on weather conditions at the project’s location. Subzone 1A shall be used for humid cities such as Jeddah, whereas subzone 1B shall be used for dry cities such as Riyadh. Additionally, zones 2 & 3 can be used for cities with mild / cold weather.
- [ ] Suitably Qualified Professional: A suitably qualified energy modeller must have a minimum of 3 years of relevant experience and a recognized qualification or chartership. The professional’s expertise should be sufficient to cover all required technical aspects to ensure the energy model is appropriate and reflects the actual performance of the building. - It is recommended that the modeler’s qualifications and experiences are carefully reviewed to ensure the professional is able to undertake modelling exercise and report accurate and reliable results for the percentage improvement in energy consumption against the baseline. Previous experience in modelling relevant types of commercial buildings will be ideal as defining geometry, zoning, and building systems performance parameters are crucial steps which require judgment and awareness of principles of buildings design considerations.
- [ ] District Cooling:
  - [ ] Buildings served by district cooling shall obtain the seasonal coefficient of performance (sCOP) from the district cooling provider. The sCOP should incorporate any thermal losses within the network.
  - [ ] If the design team is not able to obtain the sCOP from the network provider, a seasonal average COP of 4.5 must be assumed for the proposed building. The baseline building cooling system efficiency is determined using ASHRAE Standard 90.1-2013 Appendix G.
  - [ ] Baseline building shall include all equipment of both plant and demand sides. Proposed building to include demand side only along with the obtained / assumed COP.
- [ ] Renewable Energy: whilst renewable energy is separately assessed under credit E-04 Renewable Energy, any onsite renewable energy technologies (e.g. solar hot water panels, photovoltaic panels) may be accounted for in this credit by subtracting the annual renewable energy generated (whether used on-site or exported to the grid) from the total proposed annual energy consumption).

Preliminary Energy Model

The benefits of performing this exercise are:

- [ ] Provide an estimated projection of energy usage depending on various options proposed for building elements. This will support further design development activities.
- [ ] Help identify best efficiency measures of building design and performance parameters to avoid inaccurate sizing of building systems, specifically HVAC, and other aspects of architectural design of the building. Thermal Performance of the Building Envelope: Thermal conductivity (U-value) is the measurement of the flow of heat through a building material. It is defined as the rate of heat flow in watts (W) through an area of 1 square meter (m^2) for a temperature difference across the structure of 1 degree centigrade (C) or Kelvin (K). The U-value unit of measurement is therefore W/m^2K. The following procedure is recommended for a comprehensive calculation of the area-weighted U-value for the envelope’s opaque surfaces, which typically accounts for more than 50% of the envelope. It must be properly assessed to ensure the building is designed for efficiency in energy performance.
- [ ] The different sectional details of each exterior wall need to be identified at all levels and elevations. It is recommended that color coded drawings are generated to define the different types of exterior walls and its related area for all four elevations. Refer to Figure E-01.1 for guidance. Figure E-01.1 Color-coded drawing defining different types of exterior walls

- [ ] Once the different areas for wall types are identified, it is important to define sectional details of each wall type by defining the materials and related thicknesses.
- [ ] The suppliers of the proposed materials should be contacted to provide the thermal conductivity (K-value) of each type of material in order to build up the information for the area-weighted U- value calculations.
- [ ] Using the sectional details, the thermal resistance of the wall section can be calculated using the following formula: 103 ? ? ????? (?2? ??) = ???????? ????????? (?) ??????? ???????????? (? ???)
- [ ] Once the sectional R-value is identified, the U-value of the same section is calculated using the following equation (this methodology is appropriate for tall buildings with a single continuous façade system): ? ? ????? (? ?2??) = 1 ? ? ????? (?2? ??) Figure E-01.2 Example of R-value and U-value calculations
- [ ] Once the sectional U-values have been calculated, the Average Area-Weighted U-value can be calculated using the following formula: ???????? ? ? ?????(? ?2??) = ? ? x ? ? ? Figure E-01.3 Example of average area-weighted U-value calculations
- [ ] Transparent, or semitransparent, glass is not considered as the above calculation is for the opaque surfaces only.

These boxes are left blank on purpose. They list what the manual requires the plan to contain; whether the uploaded plan covers each one is for the reviewer to judge against the document itself.